Michael Liu

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Education

University of Waterloo *Sept 2025 - Apr 2030*
BCS in Computer Science, Digital Hardware Specialization

- Major Average: 93.1% | Cumulative Average: 92.7% | Student ID: 21177966
- Activities: Varsity Cross-Country, WAT.ai

Job Experience

Machine Learning Research Assistant *Toronto, ON*
The Hospital for Sick Children (SickKids) *May 2026 - Aug 2026*

- Conducted research on Reaction GFlowNets (RGFNs) for molecular generation and drug discovery.
- Designed and implemented experiments to evaluate molecular sampling strategies and generative model behavior (GitHub).

Machine Learning Scientist *Waterloo, ON*
WAT.ai *Sept 2025 - Feb 2026*

- Collaborated with a small team of students on a medical imaging project focused on improving the segmentation of microaneurysms in fundus imaging.
- Co-authored a research paper detailing our methodology and results (GitHub).

Projects

CourtHawk: Tennis Computer Vision System *github/courthawk*
OpenCV, PyTorch, MediaPipe, Pandas, scikit-learn, FastAPI, TypeScript, React

- Built an end-to-end computer vision system for tennis footage from the standard TV angle using YOLO, CNNs, and MediaPipe to track players/ball, detect court keypoints, and estimate player poses.
- Developed methods for shot detection and classification using player/ball trajectories and pose derived features.
- Projected player/ball onto a bird's-eye-view court using homography to compute movement and shot-speed statistics.
- Built and deployed a full-stack web app using FastAPI, React, and TypeScript for video upload, processing, and visualization of results.

C++ Machine Learning Library from Scratch *github/cpp-ml-from-scratch*
C++, CMake

- Built a machine learning library from scratch in C++ centered around an n-dimensional tensor implementation, neural network layers, and classical machine learning algorithms such as Linear Regression, Logistic Regression, Softmax Regression, Decision Trees/Random Forests, Naive Bayes, and KNN.
- Prioritized transparency over abstraction by keeping the math behind forward passes, backpropagation, and parameter updates explicit through the training pipeline.

Technical Skills

Programming Languages:	C/C++, Python, SQL, TypeScript
Data/Machine Learning:	PyTorch, scikit-learn, Pandas, NumPy, Matplotlib, OpenCV
Databases/Tools:	PostgreSQL, Git, GitHub, LaTeX, Excel, Tableau, CMake
Frontend:	React, HTML, CSS